Finite Color–Sign Certificates and Locality for Quadratic Strong-Operator Paving

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Abstract

We study the exact independent color–sign polynomial in the pinned manuscript on quadratic strong-operator paving over arbitrary maximal abelian subalgebras. All substantive source sections are traced, from the expectation-free reduction through invariant costs, Bernoulli coloring, nontracial moment transfer, and the final spectral cut. We do not independently establish its global theorem. Our finite results retain both random signs after color assignment and derive an even-cycle expansion, an exact sparse Hermitian realization on forests, and sharp half-radius locality for rooted moments and polynomial deletion increments of the expected polynomial itself. An explicit Bernstein construction gives dimension-independent finite-ball cost certificates and a rigorous approximate color-selection rule. A fully proved two-test selection argument reduces the assignment buffer from 80 to 32 and its cost bound from $24m$ to $9m$; with the inherited analytic and structural inputs stated explicitly, this yields a conditional paving count below $10^7\varepsilon^{-2}$. Cancellation-aware Hankel and physical-metric realizations preserve the difference between rooted atoms and full characteristic multiplicities. These use classical techniques within the precise representation contracts of the private Ram corpus, without a historical novelty or runtime claim. Contribution assessment: C3/5, exact finite-ball certificates and a smaller-buffer assignment give notable local and conditional quantitative gains while the global analytic and structural inputs remain inherited.

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1 Pinned Problem and Quantifiers

The source is OpenAI's *Quadratic Strong-Operator Paving over Arbitrary Maximal Abelian Subalgebras*, September 25, 2026, catalogue M595, family 300, pinned at

`adc7f1241b42e322a6451854ab7e4b4c146bf78a`.

The complete source-file scope and private passage locators are in Appendices A and B. The date on this report is October 7 in the specified local time zone; the execution records use October 8 UTC. Applicability L5 is not the contribution score C3.

Let H be a complex Hilbert space, $M \subseteq \mathcal{B}(H)$ a unital von Neumann algebra, and $A \subseteq M$ a unital maximal abelian $*$-subalgebra. All operators in the target theorem are bounded with domain all of H. There are no differential boundary conditions. No separability, trace, or conditional expectation onto A is assumed. A partition $P = (p_1, \dots, p_r)$ consists of orthogonal projections in A with sum one; zero pieces are allowed. Write

$$S_P(x) = \sum_{i=1}^{r} p_i x p_i.$$

The source's `thm:main` asserts the following quantifiers:

$$\forall \varepsilon \in (0,1), \qquad r = r_\varepsilon := \lceil 400000000\varepsilon^{-2} \rceil + \lceil 4/\varepsilon \rceil + 1 \le 500000000\varepsilon^{-2}, \tag{1.1}$$

followed by, for every (H, M, A, x, F, δ) with $x = x^* \in M$, $F \subset H$ finite and $\delta > 0$, the existence of P, $a = a^* \in A$ and a projection $q \in M$ such that

$$\|a\| \le \|x\|, \qquad \|(1-q)\xi\| < \delta \ (\xi \in F), \qquad \|q(S_P(x) - a)q\| \le \varepsilon \|x\|. \tag{1.2}$$

The number of colors precedes the inclusion and strong neighborhood. Neither $q \in A$ nor commutation of q with the partition is required. The strong formulation generally uses nets, not sequences. For $x = 0$ take $a = 0$, $q = 1$; otherwise scale to a contraction. Norm paving of x without compression and norm paving of strong approximants are different assertions.

The source also asserts norm paving in the Ocneanu ultrapower for an expected MASA that is countably decomposable, using Popa–Vaes' transfer theorem. Those additional hypotheses must not disappear. The source's companion approximation theorem is contextual, not an input to its quadratic proof. The completed M594 analysis concerns a different color-only polynomial, different penalties, and a sixth-power approximation-paving statement; its finite formulas cannot be copied without changes.

Status. The finite statements in Sections 3–9 are proved independently of measured coloring and arbitrary-MASA paving; the global corollary in Section 8 explicitly retains inherited analytic inputs. Section 2 audits dependencies, not all their foundations. A finite symbolic check cannot verify a theorem quantified over all nonsingular relations or all von Neumann algebras. Public literature was searched by public mathematical topic and title only; private passages remained local.

2 Stepwise Source Dependency Audit

Literal source labels below belong to the pinned source, not this report. All eight substantive section files, the main file, the bibliography and figure were read in full; there is no separate mathematical appendix.

2.1 Reduction without a normal expectation

In `sections/reductions.tex`, the source first treats the corner admitting no nonzero A-central normal positive functional. In Haagerup $L^2(M)$, decreasing diagonal pinchings of $\xi = h_\varphi^{1/2}$ tend to zero. The symmetric map

$$T_\varphi(z) = h_\varphi^{1/4} z h_\varphi^{1/4}$$

is weak-star to weak continuous, and the key block identity is

$$\sum_{e,f\in P} \|T_\varphi(ezf)\|_2^2 = \mathrm{Tr}(z^* S_P(\xi) z S_P(\xi)) \le \|z\|^2 \, \|S_P(\xi)\|_2^2 \,.$$

Two distinct randomizations are used: local selection of refinements realizing a convex approximation to a diagonal weak-star limit, then uniform coloring of a fixed fine partition. They produce a bounded self-adjoint error y with arbitrarily small symmetric seminorm and $\lceil 4/\varepsilon \rceil$ colors already fixed.

Small symmetric seminorm does not imply small $\|y\|_\varphi$. The source instead puts $\psi(z) = \varphi(y^* z y)$, uses the positive-cone inequality

$$\left\| h_\varphi^{1/2} - h_\psi^{1/2} \right\|_2^2 \le \|\varphi - \psi\| \,,$$

and chooses the support q_0 of the positive part of $\varphi - \psi$. If $u = \|T_\varphi(y)\|_2$, then $\varphi(1-q_0) + \psi(q_0) \le u$ and $\|q_0 y q_0\|_\varphi \le 3\sqrt{u}$. A spectral cut loses at most $(1 + 36/\varepsilon^2)u$ of state mass. This is a substantive standard-form argument, not a finite positive-Hankel consequence.

Let $p \in A$ be the supremum of supports of central normal positive functionals. After the finite vector tests are fixed, choose one such support $e \le p$ with $p - e$ small on them. On eMe there is a faithful central state; on $(1-p)M(1-p)$ the preceding argument applies. Neither p nor e need be central in M. Disjoint palettes remove off-corner terms under pinching; one final piece is $p - e$, and the color counts add. Faithfulness converts small state mass on projections into small mass for each fixed normal functional. The later separable reduction adjoins expectations and pinching witnesses, so that its abelian subalgebra remains a MASA, rather than merely choosing an arbitrary separable subalgebra.

2.2 Normalizer algebra and atomless kernels

With separable predual and a faithful A-central state φ, `sections/kernels.tex` proves modular invariance of the normalizer algebra N. A partial normalizer has

$$\sigma_t^\varphi(v) = v h_v^{it};$$

cutting $\log h_v$ to bounded intervals supplies entire analytic normalizers. Takesaki gives E_A, E_N preserving φ. Feldman–Moore realizes $A \subset N$ by a countable nonsingular measured relation, possibly twisted, on a standard probability space. The source uses right counting and diagonal integration, not a trace; rerooting is diagonal-unitary conjugacy.

For $y \in \ker E_N$, the joint left–right vector measure of $y\Omega$ disintegrates to the kernel of $K_y(f) = E_A(y^* f y)$. An atom would give a selected measurable graph and a nonzero bounded intertwiner $fb = b\theta(f)$ in $\ker E_N$. Its polar part is not initially a normalizer: the source passes through the type I central-support corner of $\theta(A)' \cap 1_T M 1_T$, uses its faithful central state to obtain rank-one abelian projections and partial normalizers, and then obtains $b \in N$, a contradiction. The fixed-null-set property is carefully weaker than rowwise absolute continuity. Composition preserves atomlessness by avoiding the one fixed exceptional null set of the next kernel. The atomic part of A annihilates $\ker E_N$ on both sides.

2.3 Finite polynomials and what their bounds require

The exact model in `sections/polynomials.tex` is specified in Section 3. Both signs remain independently uniform after a color is fixed, and the permanent shift changes from tW_v to $80tW_v$. The source's `lem:mixed-bounds` imports the upper bound of Ravichandran–Srivastava's Theorem 25 and their zero-matrix monotonicity. In its normalization,

$$\lambda_{\max}\chi[A_1,\dots,A_k] < 3\sqrt{2/k}, \qquad \lambda_{\max}\chi[L,-L] \ge \|L\|/2. \tag{2.1}$$

The first requires $k \ge 2$ zero-diagonal Hermitian contractions. The second uses $\chi[L,0] = \det(zI - L/2)$, retaining either L or $-L$ under monotonicity. The cited primary version contains these hypotheses [2]. The barrier theorem is not proved here.

Thus the unassigned unit-weight polynomial has largest root below $3/\sqrt{r}$. Positive principal evaluations and multiaffine Taylor expansion, not an unproved spectral congruence rule, extend zero positive cost to arbitrary positive weights when $t > 3/\sqrt{r}$. Conversely, a fully assigned block with pinched norm greater than $160t + \alpha$ has polynomial positive-root sum at least $(\alpha/2)\min W$. This lower detection bound uses the second input in (2.1); positivity of a rooted Gram matrix alone cannot replace it.

The source's assignment proof uses a common deletion spectrum Λ for all choices. With rooted mass m, old shift $a = tW_v$ and deletion increment d, it decomposes

$$d = m + e_+ + e_-, \qquad e_+ = \operatorname{Tr}(H_-)_{++}, \quad e_- = \operatorname{Tr}(H_+)_{--}.$$

The subscripts $++$ and $--$ mean deletion blocks with positive and nonpositive eigenvalues. If $m \ge a$, two Markov tests give a color with $d^i - d \le 21m$. If $m < a$, positive-block matrix inequalities give tests at $32a + \Lambda_+$ and $e_- + |\Lambda_-|$; combining side arrows with first diagonal entries $-32a$ and $-40a$ yields $d^i - d \le 7m - e_+/4$. The stated uniform result is

$$\min_i (d^i - d) \le 24m \quad \text{with buffer } \beta = 80. \tag{2.2}$$

Zero denominators are omitted only after positivity forces the corresponding residues to vanish. Nonnegative averaging then forces them to vanish in every color. Section 8 supplies the stronger buffer-32, cost-nine estimate and audits its parameter carry-through; merely replacing 24 by the visible 21 is not counted as a new contribution.

2.4 Invariant costs and uniform deletion

In `sections/invariant.tex`, compact spectral support on a fixed bounded weighted graph gives unique rooted moment limits. Deletion increments require approximation of derivatives in L^1, since their spectral-shift density is between zero and one. Uniform approximation of x_+ alone would give an error proportional to the total number of roots. Ordered changes are telescoped locally for each polynomial coefficient and then integrated using invariant mass transport; the update order is not assumed to be a well-order.

The high-weight deletion lemma additionally controls

$$J_i(v,z) = (F^0_{v,i}(z) - z - a_v)G_v(z)$$

at unassigned band vertices. A complex diagonal perturbation bounds summed deletion changes of G. Analytic left and right scaling of the off-diagonal color-i matrices, without conjugating the complex parameter, gives

$$\partial_\zeta \log P = \frac{2}{r}\sum_v \lambda_v J_i(v,z).$$

The relevant perturbation norm is at most $2|\zeta|\sqrt{K}b_2 + |\zeta|^2 b_2$ on a band $[b_1, b_2]$. Half-plane nonvanishing bounds the derivative of a logarithm with bounded imaginary part. Finite single-deletion sums pass through exhaustion by Fatou; an integrated identity first proved for local Laurent coefficients extends by analytic continuation. Constants are independent of the largest weight. That uniformity is not supplied by the bounded-ball certificates below.

2.5 Nonsingular coloring and the order of limits

In `sections/coloring.tex`, use the cocycle convention $\mu(\gamma S) = \int_S \Delta(\gamma v, v) d\mu(v)$, with edge ratios in $[K^{-1}, K]$, $K \geq 1$. The invariant lift is

$$d\nu(v, s) = d\mu(v) e^s ds, \qquad W(v, s) = e^{-s}, \qquad (v, s) \sim (u, s - \log \Delta(u, v)). \tag{2.3}$$

This convention is the sign reversal of a common height convention and must be used consistently. A weight band $l \leq W \leq H$ has finite invariant measure $l^{-1} - H^{-1}$. The high-cutoff deletion estimate controls a set of measure H^{-1}; with a fixed high cutoff, edge ratios make a sufficiently low cutoff invisible on finite balls. Transform comparisons, second-moment uniform integrability, and Poisson smoothing of the spectral shift produce joint band L^1 limits. Rational height-translation invariance makes $m/W, d/W, (d^i)/W$ independent of height. The proof needs uniformity over configurations and auxiliary probability extensions.

Bad fully assigned balls are separated into finitely many center classes at distance greater than $2m+1$. Deletion monotonicity, invariant mass transport, and (2.1) then bound their base probability by $C_4\mathcal{S}$, where $\mathcal{S} = \int \bar{m}\, d\mu$. Fresh uniform times select a small fraction h of unassigned vertices. Color choices are determined before those times. The hazard identity is proved on finite strips, conditioning on every time except the root time, before choosing homogeneous limit versions. Its survival factor $(1-\theta)^{-1}$ cancels the independent old-cost survival probability. Consequently

$$\mathcal{S}(c^h) \leq (1 + 24h)\mathcal{S}(c) + hg(h), \qquad g(h) \longrightarrow 0.$$

For $N = \lceil T_0/h \rceil$ the cost tends to zero at fixed T_0, while a prescribed ball is assigned with failure probability at most $D_m e^{-T_0}$. Uniform final colors complete the rest. Permutation-covariant random tie-breaking proves conditional color marginals exactly $1/r$. A deterministic least-color tie-break is not a substitute for that symmetry.

2.6 One partition and nontracial moment transfer

In `sections/transfer.tex`, the relative Bernoulli algebra $\widetilde{N}$ has a faithful normal expectation onto N. The reduced amalgamated product $\mathcal{K} = M *_N \widetilde{N}$ and its faithful state Φ put the label algebra B in the centralizer. If $p \in B$ and $\widetilde{E}_N(p) = \lambda 1$, the reduced-word projection gives

$$\|pwp\| \leq (\lambda + 2\sqrt{\lambda})\, \|w\| \qquad (w \in \ker E_N).$$

Thus the same color projections used for the Cartan component control the complementary component, without a partition product.

Finite-symbol cylinder substitution into a fine partition of A respects identities because every assignment to finitely many distinct sites has positive conditional probability. It need not preserve conditional centering. Fixed scalar moment transfer therefore expands and subtracts model expectations, approximates centered letters by modular-entire ones, and uses KMS rotation rather than trace cyclicity. Fourier exponent cancellation yields equality patterns. A shortest cyclic pair and an auxiliary-phase forest isolate a small endpoint collision while bounding the other Cauchy–Schwarz

square sum. Interior pinchings converge with coarse endpoints fixed; only afterward are the endpoints refined, where the atomless composed kernels annihilate the graph collision. Interchanging these limits or replacing nonsingular partial maps by measure-preserving ones is unjustified. The included figure depicts selected links in this proof, not an additional theorem.

2.7 Clipping, fixed moments, and corner assembly

In `sections/completion.tex`, write $x - E_A x = y + w$, with $y = E_N x - E_A x$, $w = x - E_N x$, both of norm at most two for a contraction x. Finite-support y_n remain bounded by two after diagonal removal and bounded-cocycle compression. Choose $r = \lceil 20000^2 \varepsilon^{-2} \rceil$, $t = 4/\sqrt{r}$, and apply coloring to $y_n/2$ with radius n, $\alpha = \varepsilon/32$, and failure probability $(\varepsilon/16)^{2n}$. Its Cartan bound is

$$1280/\sqrt{r} + \varepsilon/16 \le 253\varepsilon/2000 < \varepsilon/4;$$

the free part is bounded by $2/r + 4/\sqrt{r} < \varepsilon/4$.

The high moments concern pinched y_n, not the original y. Clipping at $\varepsilon/2$ is transferred to y using an algebra of varying right multipliers that preserve bounded state-null sequences. Centralizer sequences and constant sequences belong to it; arbitrary bounded varying sequences do not. Only after clipping does the proof fix a moment $2k$, then choose n, then a finite cylinder approximation, then a fine base partition and a deterministic evaluation. The final cut $q = 1_{[-\varepsilon,\varepsilon]}(z)$ follows from $\varphi(z^{2k}) < \rho\varepsilon^{2k}$. No rate for $y_n \to y$ exponentially fast in k is assumed. Adding the expectation-free palette and discarded-corner piece gives (1.1).

Unmet verification boundary. This chain retains the imported mixed-root bounds, Takesaki expectations, Feldman–Moore representation, standard-form and type I structure, and reduced amalgamated-product foundations. It also retains the source's uniform analytic cutoff and nontracial transfer arguments. Reading their proofs and testing finite identities is not an independent proof of the entire chain. No source contradiction or global theorem is asserted here.

3 The Exact Finite Color–Sign Model

Let I be finite, $T = T^* \in \mathbb{C}^{I\times I}$, $T_{vv} = 0$, $\|T\| \le 1$, $r \in \mathbb{N}$, and $t > 0$. The pinned source uses $\beta = 80$; the finite identities below hold for every fixed $\beta \ge 1$, and Section 8 proves the assignment estimate with $\beta = 32$. Choose positive weights W_v, a partial coloring $c : I \to \{0, 1, \dots, r\}$, and set

$$a_v = \tau_v W_v, \qquad \tau_v = t \ (c(v) = 0), \quad \tau_v = \beta t \ (c(v) > 0).$$

At v, choose a color with law

$$p_v(i) = \begin{cases} 1/r, & c(v) = 0, \\ \mathbf{1}_{\{i=c(v)\}}, & c(v) > 0, \end{cases} \tag{3.1}$$

and an independent uniform sign in $\{+1, -1\}$, independently across all sites. Denote the pair by $\omega_v = (i_v, \sigma_v)$. Put $C = D_W T D_W$, $D_W = \operatorname{diag}(\sqrt{W_v})$. The outcome B_ω has diagonal $-a_v$ and, for $u \ne v$,

$$(B_\omega)_{uv} = \begin{cases} \sigma_u C_{uv}, & \omega_u = \omega_v, \\ 0, & \omega_u \ne \omega_v. \end{cases} \tag{3.2}$$

This is not the matrix obtained by multiplying C_{uv} by $\sigma_u\sigma_v$, nor the ordinary color pinching of T.

For $Z = \operatorname{diag}(z_v)$ define $P_I(Z) = \mathbb{E}\det(Z - B_\omega)$, $P_I(z) = P_I(z, \dots, z)$, and $P_\varnothing = 1$. Restriction always means principal restriction retaining the laws and penalties at the surviving sites. Put $Q_v = P_{I\setminus\{v\}}$. At an unassigned v, $P^0_{v,i}$ fixes its color i but leaves both signs and its shift $a_v = tW_v$; $P_{v,i}$ also replaces a_v by βa_v. In particular, all these choices have the same Q_v.

Proposition 3.1 (Differentiation and positivity)**.** *Let $\mathcal{A}_v$ be the allowed pair set, $k_v = |\mathcal{A}_v| \in \{2, 2r\}$, $U_j = \{v : j \in \mathcal{A}_v\}$, and $B_j = \sigma C - \operatorname{diag}(a)$ for $j = (i, \sigma)$. Then*

$$P_I(Z) = \left(\prod_{v\in I} \frac{\partial_{z_v}^{k_v-1}}{k_v!}\right) \prod_j \det((Z - B_j)|_{U_j}), \qquad \partial_{z_v} P_I(Z) = P_{-v}(Z|_{-v}). \tag{3.3}$$

The polynomial and all principal restrictions are multiaffine and upper-half-plane stable. For $z \in \mathbb{C}^+$,

$$F_v(z) := \frac{P_I(z)}{Q_v(z)} = z + a_v - \sum_\lambda \frac{b_{v,\lambda}}{z - \lambda}, \quad b_{v,\lambda} \geq 0, \qquad \Im F_v(z) \geq \Im z. \tag{3.4}$$

The sum ranges over distinct roots of Q_v, allowing zero residues.

Proof. Each determinant is affine in z_v. Differentiating $k_v - 1$ times leaves v in just one factor, with multiplier $(k_v - 1)!/k_v! = 1/k_v$; terms in which it remains in more than one factor are impossible after that many derivatives. Carrying this out at every vertex gives the independent outcome average. Differentiating the resulting multiaffine determinant average once more deletes the vertex, proving the second identity.

Every factor is nonzero when all diagonal imaginary parts are positive, since a null vector of $Z - B_j$ would contradict its strictly positive imaginary quadratic form. Products preserve nonvanishing. Univariate Gauss–Lucas preserves this property under differentiation; the possibility of an identically zero derivative is excluded by induction over processed vertices and their principal restrictions, because the leading coefficient in an unprocessed variable is the already-processed expression with that site deleted. This proves stability, and conjugation implies real-rootedness after uniform specialization.

Fix the other variables in the upper half-plane. The quotient by the deletion polynomial is monic affine in z_v, so its only zero has imaginary part at most zero. At uniform specialization this says $F_v - z$ has nonnegative imaginary part. Its real rational poles must be simple with nonpositive residues: a pole of higher order would change the sign of the imaginary part on upper rays, and vertical approach fixes the sign at a simple pole. The constant term at infinity is a_v, from the deterministic diagonal. This proves (3.4), including canceled poles. □

Proposition 3.2 (Full and rooted spectral records)**.** *Let Λ_v list all roots of Q_v with multiplicity, and put one coordinate of squared magnitude $b_{v,\lambda}$ at one copy of each distinct root, with zero at its other copies. The resulting vector g_v defines*

$$H_v = \begin{pmatrix} -a_v & g_v^* \\ g_v & \Lambda_v \end{pmatrix}, \qquad \det(zI - H_v) = P_I(z), \qquad G_v(z) := Q_v/P_I = e_0^*(zI - H_v)^{-1}e_0. \tag{3.5}$$

Its root spectral measure ρ_v has mass one, and

$$G_v(z) = \int \frac{\rho_v(dx)}{z - x}, \qquad \sum_v \rho_v = \sum_{P_I(x)=0} \delta_x. \tag{3.6}$$

All roots on the right are counted with multiplicity. If

$$s(P) = \sum_{P(x)=0} x_+, \quad m_v = \int x_+ \rho_v(dx), \quad d_v = s(P_I) - s(Q_v),$$

then $\sum_v m_v = s(P_I)$ *and* $0 \le d_v$. *Also* $\frac{1}{r}\sum_i F^0_{v,i} = F_v$ *and* $F_{v,i} = F^0_{v,i} + (\beta - 1)a_v$.

Proof. The Schur complement in (3.5) is (3.4); this proves the determinant and resolvent identities. The finite spectral theorem gives the measure and its unit mass. Summing the multivariate deletion identity on the diagonal gives $\sum_v G_v = P_I'/P_I$, the Cauchy transform of the full root-counting measure. Uniqueness of rational partial fractions proves (3.6). Principal-matrix interlacing of H_v and Λ_v gives $0 \le n_{P_I}(x) - n_{Q_v}(x) \le 1$, where $n_P(x)$ counts roots strictly greater than x. Integration over $(0, \infty)$ proves $d_v \ge 0$. Conditioning the root color averages back to its old law, and changing only that diagonal adds $(\beta - 1)a_v Q_v$, proving the last identities. □

Certified support. If all $W_v \le H_0$, every outcome has norm at most $L = (1+\beta t)H_0$: compressions and pinching of a contraction are contractions. For real $z > L$ all determinants are positive; for $z < -L$ all have sign $(-1)^{|I|}$. Hence their real-rooted average and every restriction or root option have roots in $[-L, L]$. Rooted measures have that support as well. There is no bound uniform in H_0 here.

4 Even Cycles and a Sparse Forest Realization

For a directed cycle $\gamma = (v_1, \dots, v_\ell)$ with distinct vertices and $\ell \ge 2$, taken modulo cyclic rotation but not reversal, put

$$\kappa(\gamma) = \begin{cases} 2^{1-\ell} \displaystyle\sum_{i=1}^{r} \prod_{v \in \gamma} p_v(i), & \ell \text{ even}, \\ 0, & \ell \text{ odd}. \end{cases} \tag{4.1}$$

An orientation reversal is a separate cycle for length at least three; the transposition is counted only once.

Theorem 4.1 (Exact signed cycle expansion)**.** *For all the finite data in Section 3,*

$$P_I(Z) = \sum_{\mathcal{C}} \prod_{v \notin V(\mathcal{C})} (z_v + a_v) \prod_{\gamma \in \mathcal{C}} \left[-\kappa(\gamma) \prod_{j=1}^{|\gamma|} C_{v_j v_{j+1}} \right], \tag{4.2}$$

where $v_{|\gamma|+1} = v_1$ *and the sum runs over collections of pairwise vertex-disjoint directed cycles of length at least two, including the empty collection. Odd cycles contribute zero. If a cycle has conflicting assigned colors its weight is zero; if it has* u *unassigned sites and at least one assigned site all of the same color, its even-cycle weight is* $2^{1-\ell} r^{-u}$. *An all-unassigned even cycle has weight* $(2r)^{1-\ell}$.

Proof. Expand the determinant into permutations. Fixed points give $z_v + a_v$. A nontrivial cycle contributes a permutation sign $(-1)^{\ell-1}$ and ℓ negative off-diagonal entries, hence a combined minus sign. Its edges survive exactly when every site in the cycle has the same color and sign. For a fixed color i and sign σ, this event has probability $2^{-\ell} \prod_{v \in \gamma} p_v(i)$ and contributes σ^ℓ. Summing over $\sigma = \pm 1$ gives (4.1). Disjoint cycles depend on disjoint site labels and therefore multiply. The remaining formulas follow from (3.1). This also proves the expansion for arbitrary independent color laws p_v summing to one, as a finite algebraic statement; no new general root bound for such laws is asserted. □

In particular, the unshifted polynomial has only powers of parity $|I|$, even with unequal weights and partial colors. Equal deterministic shifts $a_v = a$ translate this symmetry to the point $-a$. Unequal shifts do not imply symmetry around a single point. A vertex-sign conjugation would retain odd cycle products, so it is not an interchangeable model.

Theorem 4.2 (Exact forest realization)**.** *Suppose the support graph of T is a finite forest. Define*

$$q_{uv} = \frac{1}{2}\sum_i p_u(i)p_v(i), \qquad K_{uv} = \sqrt{q_{uv}}\,C_{uv}\ (u \neq v), \quad K_{vv} = -a_v. \tag{4.3}$$

Then $K = K^*$ and, for every principal set $J \subseteq I$,

$$P_J(Z) = \det(Z|_J - K|_J), \qquad G_v(z) = ((zI-K)^{-1})_{vv}, \qquad \rho_v = \text{the spectral measure of } K \text{ at } e_v. \tag{4.4}$$

The identities include the conditioned and permanently assigned root options, using their own q_{uv} and a_v.

Proof. On a forest the only nonzero permutation cycles are transpositions: a cycle with at least three distinct vertices would give an undirected cycle in the support graph. Thus (4.2) is a matching expansion, and every edge contributes $-q_{uv}|C_{uv}|^2$. The determinant of $Z - K$ has exactly the same matching expansion because $|K_{uv}|^2 = q_{uv}|C_{uv}|^2$. Restrictions remain forests and retain these factors, so the identity holds simultaneously for all principal sets. Cramer's rule gives the resolvent formula and hence the spectral-measure formula. Root conditioning only changes incident edge factors and the root diagonal, preserving the argument. □

This gives an actual sparse Hermitian operator for the expected polynomial on forests, rather than the dense, root-dependent arrow. It is not the expectation of B_ω: expectation multiplies an off-diagonal entry by a signed probability, which is zero. It is not generally valid on a graph with cycles. For example, on a triangle the odd cycle is absent from P but present in the characteristic polynomial of the positive-edge matrix in (4.3). On a four-cycle the source cycle coefficient and the product of edge square roots also need not agree. Thus the theorem's acyclicity assumption has real content.

5 Sharp Half-Radius Mixed-Polynomial Locality

Fix an undirected support graph and measure distance in it, retaining that enclosing graph when vertices are deleted. The following result concerns Q_v/P_I, not an expected random-matrix resolvent. It therefore needs a separate proof from the usual closed-walk argument for a fixed matrix.

For $j \geq 0$, let $\mu_j(v) = \int x^j \rho_v(dx)$. For a real polynomial f, define

$$d_v[f] = \sum_{P_I(x)=0} f(x) - \sum_{Q_v(x)=0} f(x).$$

The constants have $\mu_0(v) = 1$ and $d_v[1] = 1$.

Theorem 5.1 (Mixed moments and increments on a half-radius ball)**.** *For every integer $k \geq 0$, the complete finite data on $B(v, \lfloor k/2 \rfloor)$ determine all $\mu_j(v)$ and $d_v[x^j]$ for $0 \leq j \leq k$. Principal restriction to that ball leaves those quantities unchanged. The statement holds with arbitrary deterministic diagonal shifts and for either root color option. The coefficient of z^{-j} in $F_v(z) - z - a_v$, $j \geq 1$, is determined by radius $\lfloor (j+1)/2 \rfloor$. In general the radius for moments through degree $k \geq 2$ cannot be reduced by one.*

Proof. Put $u_v = 1/z_v$ and $\mathcal{H}_I(u) = P_I(Z)\prod_v u_v$, with constant term one. Give each diagonal parameter and each directed edge entry an independent formal tag before taking the determinant average. A term of $\mathcal{H}_I$ is a union of tagged directed permutation cycles and diagonal monomers. A length-ℓ off-diagonal cycle has ℓ edge tags and u-degree ℓ; a monomer has u-degree one. This description follows directly from the permutation expansion, with the scalar label averages absorbed into coefficients. Products of such terms have a balanced directed edge multigraph: every vertex has equal entering and leaving multiplicity.

Expand the formal logarithm

$$\log \mathcal{H}_I = \sum_{s\ge 1} \frac{(-1)^{s+1}}{s}(\mathcal{H}_I - 1)^s.$$

At any fixed u-degree this is a finite calculation. A surviving tagged monomial has connected support in its used-edge multigraph, with a lone diagonal vertex allowed. To see the necessary cancellation rigorously, fix its used edges and set all other edge tags to zero. This cannot alter its coefficient. If its support splits between disconnected components of those edges, the original determinant average factors over those components, since the site labels are independent. Its formal logarithm is the sum of their logarithms and has no monomial spanning two components. Thus that coefficient is zero. The same argument includes diagonal tags at isolated vertices. Tag specialization afterward preserves every conclusion.

Consider a nonzero degree-j term of this logarithm whose support contains v. If there are edges, their balanced connected multigraph admits a closed Euler tour; the number of edges with multiplicity is at most j, since diagonal monomers also consume degree. A vertex at graph distance h from v requires at least h edges on a route out and at least h on its return in that closed tour. Hence $2h \le j$. All its data lie in $B(v, \lfloor j/2 \rfloor)$. With no edges the term depends only on v.

Now

$$G_v(Z) = \partial_{z_v} \log P_I(Z) = u_v - u_v^2 \partial_{u_v} \log \mathcal{H}_I(u). \tag{5.1}$$

The derivative selects precisely the terms containing v and increases the total u-degree by one. After uniform specialization, the coefficient of z^{-1-j} is $\mu_j(v)$ by the Cauchy transform. This proves its radius claim. To obtain deletion increments without doubling a radius, use instead

$$\log F_v(z) = \log z - \sum_{j\ge 1} \frac{d_v[x^j]}{j z^j} = \log z + \log \mathcal{H}_I(z^{-1}) - \log \mathcal{H}_{-v}(z^{-1}). \tag{5.2}$$

The formal difference selects exactly the tagged monomials containing v, because deleting a site is setting its u variable to zero before specialization. The same degree argument proves the claim for $d_v[x^j]$. Constants were dealt with separately. Formal inversion of $G_v = z^{-1}(1 + \sum_{j\ge 1} \mu_j z^{-j})$ uses only $\mu_0, \dots, \mu_{j+1}$ for the coefficient of z^{-j} in F_v, proving its assertion. Conditioning or changing one diagonal preserves the cycle and independence arguments.

For sharpness set $h = \lfloor k/2 \rfloor \ge 1$, take the path $0, \dots, h$ with edge entries $1/3$, weights one, all sites assigned color one, buffer $\beta = 80$, and $t = 1/80$. Thus $a_v = 1$ and T is a contraction, since its absolute row sums are at most $2/3$. Theorem 4.2 gives $K = -I$ plus path edges $1/(3\sqrt{2})$. At root zero and degree $2h$, the unique closed walk reaching h and returning has weight

$$(1/18)^h > 0. \tag{5.3}$$

All other walks remain in the radius-$(h-1)$ ball; diagonal steps cannot occur in a length-$2h$ walk that reaches h. Thus deleting h changes $\mu_{2h}(0)$ by exactly (5.3). This is a source-admissible color–sign example, not merely a deterministic-matrix analogy. □

Relation to the source. The source's `lem:rooted-locality` uses radius k for a degree-k rooted moment, by connected support alone. The extra balanced-edge argument proves radius $\lfloor k/2 \rfloor$ for the same expected polynomial, and (5.2) gives that radius directly for polynomial deletion increments. The previous M594 report established half-radius locality only in a later deterministic descent step; it explicitly left its mixed-polynomial locality unchanged. The present theorem is a separate mixed-polynomial proof. No claim of historical priority is made.

If the graph has degree at most $D \geq 2$, the ball bound is

$$|B(v,h)| \leq 1 + D\sum_{j=0}^{h-1}(D-1)^j.$$

For $D \geq 3$ the sufficient worst-case neighborhood bound changes exponentially as a function of polynomial degree. This is an information-radius reduction, not a measured computational speedup. Calculating a determinant on that ball can still be expensive.

6 Explicit Finite-Ball Cost Certificates

Assume henceforth the bounded-weight support certificate $[-L, L]$, $L = (1+\beta t)H_0 > 0$, simultaneously for the original configuration, all deletions, and all root options. For an integer $N \geq 2$, define

$$B_N(x) = \sum_{j=0}^{N}(2Lj/N - L)_+ \binom{N}{j} s^j(1-s)^{N-j}, \qquad s = \frac{x+L}{2L}. \tag{6.1}$$

This is an explicit degree-at-most-N polynomial, not a sampled approximation to a nonsmooth function.

Lemma 6.1 (Two different approximation norms)**.** *On $[-L, L]$,*

$$0 \leq B_N(x) - x_+ \leq L/\sqrt{N}, \qquad B_N(-L) = 0, \qquad \int_{-L}^{L} |B_N'(x) - \mathbf{1}_{(0,\infty)}(x)|dx \leq L/\sqrt{N}. \tag{6.2}$$

Proof. For $J \sim \text{Binomial}(N, s)$, $B_N(x) = \mathbb{E}(2LJ/N - L)_+$. Jensen gives the lower bound. The scalar positive-part function is 1-Lipschitz, so the upper bound is

$$2L\mathbb{E}|J/N - s| \leq 2L\sqrt{s(1-s)/N} \leq L/\sqrt{N}.$$

The endpoint identities are $B_N(-L) = 0$, $B_N(L) = L$. Differentiating a Bernstein polynomial expresses B_N' as a convex combination of the discrete slopes

$$\frac{N}{2L}[(2L(j+1)/N - L)_+ - (2Lj/N - L)_+],$$

which belong to $[0, 1]$. Hence the derivative error integral is exactly $2B_N(0)$. At $s = 1/2$ the random variable $X = 2LJ/N - L$ is symmetric with mean zero, so $2B_N(0) = \mathbb{E}|X| \leq \sqrt{\mathbb{E}X^2} = L/\sqrt{N}$. This proves the derivative bound separately from the uniform bound. □

Theorem 6.2 (Bounded-cut local certification)**.** *Let $R = \lfloor N/2 \rfloor$ and compute the finite polynomial pair on $B(v, R)$, retaining the original data and any current vertex deletions. Its exact polynomial moments give*

$$M_N(v) = \int B_N(x)\rho_v(dx), \qquad D_N(v) = d_v[B_N]$$

for the full finite graph, and, when the graph is infinite with bounded degree, for its exhaustion-independent rooted limits. The same ball computes all $D_N^i(v) = d_v^i[B_N]$ at an unassigned root. With $e_N = L/\sqrt{N}$,

$$\max(0, M_N - e_N) \le m_v \le M_N, \qquad \max(0, D_N - e_N) \le d_v \le D_N + e_N, \tag{6.3}$$

and $|D_N^i - d_v^i| \le e_N$ for every color. Consequently a prescribed absolute error $\delta > 0$ is obtained by choosing

$$N \ge \max\{2, \lceil (L/\delta)^2 \rceil\}, \qquad R = \lfloor N/2 \rfloor. \tag{6.4}$$

These estimates are independent of the number of vertices.

Proof. Theorem 5.1 makes every polynomial quantity exactly local with the stated radius, including deletion increments. Uniform function error in Lemma 6.1 gives the rooted bound because ρ_v is a probability measure. For the increment use the spectral shift $\xi_v(x) = n_{P_I}(x) - n_{Q_v}(x)$, $0 \le \xi_v \le 1$. Every absolutely continuous f satisfies

$$d_v[f] = f(-L) + \int_{-L}^{L} f'(x)\xi_v(x)dx.$$

Indeed write $f(\lambda) = f(-L) + \int_{-L}^{\lambda} f'(x)dx$ for each root and subtract, noting the degree difference one. Thus the derivative bound, together with $B_N(-L) = (-L)_+ = 0$, gives the dimension-free increment error. The same argument applies separately to each option.

On a bounded-degree infinite graph, balls are finite. Each polynomial moment and increment is eventually fixed along an exhaustion by Theorem 5.1. Probability measures on $[-L, L]$ are compact and determined by polynomial integrals, yielding unique rooted limits. The displayed uniform increment errors show that the increments themselves are Cauchy: choose N to make two approximation errors small and then include its ball. The same estimates pass to the limits. This establishes the assertion without any unbounded-weight limit. □

The explicit radius replaces a qualitative bounded-cut polynomial-approximation step in the source's `lem:bounded-costs`. It does not make the homogeneous-cutoff limit effective, since L grows with the largest retained weight. Taking $H_0 \to \infty$ in (6.4) gives no useful fixed radius.

Corollary 6.3 (Certified approximate color selection)**.** *Suppose* $\min_i(d_v^i - d_v) \le Cm_v$. *The pinned source supplies $C = 24$, $\beta = 80$; Theorem 8.2 below proves $C = 9$ for every $\beta \ge 32$. At an unassigned v, choose a color minimizing the locally computed $D_N^i(v)$. Then*

$$d_v^{i_*} - d_v \le Cm_v + 2e_N. \tag{6.5}$$

If computed values $\widehat{D}_N^i$ carry certified absolute arithmetic error at most η, minimizing them gives $Cm_v + 2(e_N + \eta)$ instead. A posteriori, a particular choice satisfies the exact inequality if certified bounds $d_v^i \le U_i$, $d_v \ge L_d$, $m_v \ge L_m$ obey $U_i \le L_d + CL_m$.

Proof. Let j be an exact minimizing color. Then $d_v^{i_*} \le D_N^{i_*} + e_N \le D_N^j + e_N \le d_v^j + 2e_N$. Insert the assumed exact assignment inequality. The same inequalities with the two arithmetic errors prove the second statement. The last assertion follows by substituting lower bounds for the unassigned configuration; upper bounds would not certify it. □

All minimizers satisfy the bound. To preserve the conditional uniform marginals needed in the source, choose the first minimizer in a fresh uniform random permutation of color names. With exact polynomial evaluation, the local quantities are permutation covariant, so this rule is equivariant and its output is conditionally uniform whenever the input construction is equivariant. A floating-point implementation with label-dependent rounding needs its own symmetry account. No infinite-step coloring is certified by this corollary alone.

7 Sign-Aware Moment and Mass Budgets

The source bounds collision probabilities by one. Here the sign law gives a sharper exact quantity:

$$q_{uv} = \mathbb{P}(\omega_u = \omega_v) = \begin{cases} 1/(2r), & \text{at least one of } u, v \text{ is unassigned,} \\ 1/2, & \text{both assigned the same color,} \\ 0, & \text{assigned distinct colors.} \end{cases} \tag{7.1}$$

Define $S_v = \sum_{u \neq v} W_v W_u |T_{vu}|^2 q_{uv}$. Expansion at infinity gives

$$\mu_0 = 1, \qquad \mu_1 = -a_v, \qquad \mu_2 = a_v^2 + S_v, \qquad \sum_{\lambda} b_{v,\lambda} = S_v. \tag{7.2}$$

This follows either from the singleton and transposition terms of Theorem 4.1, or by the first two powers of the arrow in (3.5).

Proposition 7.1 (Local positive-cost budget)**.** *For every source configuration and root option,*

$$m_v \leq \frac{\sqrt{a_v^2 + S_v} - a_v}{2} = \frac{S_v}{2(\sqrt{a_v^2 + S_v} + a_v)}. \tag{7.3}$$

If neighboring weight ratios lie in $[K^{-1}, K]$, then $S_v \leq K W_v^2/2$ for every configuration, and $S_v \leq K W_v^2/(2r)$ at an unassigned root before conditioning. Thus there

$$\frac{m_v}{W_v} \leq \frac{\sqrt{t^2 + K/(2r)} - t}{2}.$$

At an unassigned v conditioned on color i, the precise residue mass is

$$S_v^i = \frac{W_v}{2} \left[\sum_{u : c(u) = i} W_u |T_{vu}|^2 + \frac{1}{r} \sum_{u : c(u) = 0,\ u \neq v} W_u |T_{vu}|^2 \right]. \tag{7.4}$$

The permanent option has the same S_v^i and shift $\beta t W_v$.

Proof. Since $x_+ = (|x| + x)/2$, Cauchy–Schwarz and (7.2) give $m_v = (\int |x| d\rho_v - a_v)/2 \leq (\sqrt{a_v^2 + S_v} - a_v)/2$. Rationalizing gives the other form. The collision table and $\sum_u |T_{vu}|^2 \leq 1$ prove the ratio bounds. Conditioning fixes only the root color; an assigned neighbor of that color collides in sign with probability $1/2$, and an unassigned neighbor collides in pair with probability $1/(2r)$. This proves (7.4). □

These are valid small local budgets, but an upper bound for m_v must not be used as the lower bound in the final acceptance test of Corollary 6.3. They sharpen integrable domination and diagnostics, not the imported global root bound. They are not alone the basis of the contribution rating.

8 A Smaller Buffer and Its Conditional Global Consequence

This section proves a finite improvement of the source's actual assignment step, not just a proxy norm. It then separates the numerical carry-through from the inherited analytic arguments.

Lemma 8.1 (Two tests at the boundary factor two)**.** *Let $A_i, B_i \ge 0$ for $1 \le i \le r < \infty$, and let $\bar{A}, \bar{B}$ denote their uniform averages. There is an i with $A_i \le 2\bar{A}$ and $B_i \le 2\bar{B}$.*

Proof. If $\bar{A} = 0$, all $A_i = 0$ and there are no bad indices for the first test. If $\bar{A} > 0$, the set $E_A = \{i : A_i > 2\bar{A}\}$ has size strictly less than $r/2$: otherwise $\sum_{i \in E_A} A_i > 2\bar{A}|E_A| \ge r\bar{A}$, a contradiction. The empty set also has size less than $r/2$. The same reasoning applies to E_B. Thus their union has size less than r, leaving a common good index. Strictness belongs to the bad event, so no artificial factor $2 + \eta$ is required. □

Theorem 8.2 (Buffer 32 and assignment cost nine)**.** *Let $a > 0$, let Λ be a finite real diagonal matrix, and let the nonnegative residues b_λ be the uniform averages of $b_{i,\lambda} \ge 0$, $1 \le i \le r$. On one copy of each repeated pole set the arrow coupling to $\sqrt{b_\lambda}$, and use zero on its other copies. Define*

$$H = \begin{pmatrix} -a & g^* \\ g & \Lambda \end{pmatrix}, \qquad H_i(\beta) = \begin{pmatrix} -\beta a & g_i^* \\ g_i & \Lambda \end{pmatrix}, \quad m = (H_+)_{00}, \quad d = \operatorname{Tr} H_+ - \operatorname{Tr} \Lambda_+.$$

For every $\beta \ge 32$, some i satisfies

$$\operatorname{Tr}(H_i(\beta))_+ - \operatorname{Tr} \Lambda_+ - d \le 9m. \tag{8.1}$$

For $m < a$ the right side can be replaced by $3m$. Empty blocks, repeated poles, zero residues, and zero eigenvalues are allowed. In particular the exact source color–sign model satisfies $d_v^i - d_v \le 9m_v$ with buffer 32, and also with its original buffer 80.

Proof. Split the deletion coordinates into $\lambda > 0$ and $\lambda \le 0$. Write $H = H_+ - H_-$ and put $e_+ = \operatorname{Tr}(H_-)_{++}$, $e_- = \operatorname{Tr}(H_+)_{--}$. Diagonal block comparison gives

$$d = m + e_+ + e_-, \qquad (H_-)_{00} = m + a. \tag{8.2}$$

We also need $d \le 2m + a$. Let $E = e_0 e_0^*$, $D = 1 - E$ and $R = 1_{(0,\infty)}(H)$. Since DRD is a positive contraction on the deletion space, the variational formula gives $\operatorname{Tr} \Lambda_+ \ge \operatorname{Tr}(\Lambda DRD)$. Therefore

$$d \le \operatorname{Tr}((EH + HE - EHE)R) = 2m + a\langle e_0, Re_0 \rangle \le 2m + a.$$

Suppose first that $m \ge a$. Retain only the positive deletion block and root, with root diagonal $-a$. Its unique negative eigenvalue is $-L_0$, where

$$L_0 = a + \sum_{\lambda > 0} \frac{b_\lambda}{L_0 + \lambda}, \qquad d^+ = L_0 - a.$$

An empty block gives $L_0 = a$, $d^+ = 0$. For the nonpositive deletion block let $s_0 \ge 0$ be its single positive eigenvalue if one exists and zero otherwise, so $d^- = s_0$ and

$$\sum_{\lambda \le 0} \frac{b_\lambda}{s_0 + |\lambda|} \le s_0 + a.$$

When $s_0 = 0$, all residues at zero must vanish, since a positive residue there forces a positive zero of the quotient; omit those terms. If the remaining sum at zero exceeded a, continuity of its decreasing sum would again force a positive zero. This proves the inequality at that endpoint. Pinching off the unused root coupling cannot increase positive trace, hence $d^+, d^- \le d$.

Apply Lemma 8.1 to the two color-conditioned residue sums at denominators $L_0 + \lambda$ and $s_0 + |\lambda|$. Their means are at most d^+ and $s_0 + a$, respectively. Some color thus has tests at most $2d^+$ and

$2(s_0 + a)$. All omitted zero residues also vanish in each color by nonnegative averaging. For this color the positive-block arrow with root diagonal $-a$ has increment at most $2d^+$: if its negative root magnitude is below L_0 the claim is immediate, and otherwise its root equation is bounded by the test at L_0. The nonpositive-block arrow with root diagonal $-2a$ has increment at most $2s_0$, since a positive root $s > 2s_0$ would require residue sum $s + 2a > 2(s_0 + a)$, while its decreasing sum is bounded by the test at s_0.

Extend these side arrows by zero on their complementary blocks and add. Their root diagonal is $-3a$. Subadditivity of positive trace gives its increment at most $2(d^+ + d^-) \le 4d$. Decreasing the root diagonal to $-\beta a$, $\beta \ge 3$, cannot increase the positive trace. Thus the increase over d is at most $3d \le 3(2m + a) \le 9m$.

Now suppose $m < a$. Positivity of H_+ and H_- implies that the outer product of a coupling column is bounded by the root diagonal times the corresponding block. Apply this to both parts and use $(p - q)(p - q)^* \preceq 2pp^* + 2qq^*$ to obtain

$$g_+ g_+^* \preceq 2m\Lambda_+ + (4m + 2a)(H_-)_{++}, \tag{8.3}$$

$$g_- g_-^* \preceq 2(m + a)|\Lambda_-| + (4m + 2a)(H_+)_{--}. \tag{8.4}$$

Conjugate (8.3) by $(12a + \Lambda_+)^{-1/2}$. The left side is rank one, so its norm is the residue sum. Bound the error term by its trace and use $4m + 2a \le 6a$:

$$\sum_{\lambda > 0} \frac{b_\lambda}{12a + \lambda} \le 2m + \frac{1}{2} e_+.$$

Put $L_1 = e_-$. If $L_1 > 0$, conjugating (8.4) by $(L_1 + |\Lambda_-|)^{-1/2}$ gives

$$\sum_{\lambda \le 0} \frac{b_\lambda}{L_1 + |\lambda|} \le 2(m + a) + (4m + 2a) \le 10a,$$

because the conjugated positive error has norm at most $\mathrm{Tr}(H_+)_{--}/L_1 = 1$. If $L_1 = 0$, positivity forces $(H_+)_{--} = 0$. Equation (8.4) then forces zero coupling on $\ker |\Lambda_-|$; use the inverse on its support only. The same upper bound holds with these zero terms omitted.

The two-test lemma gives a color whose positive test is at most $4m + e_+$ and negative test at most $20a$. Its positive-block arrow with root diagonal $-12a$ has increment at most $4m + e_+$, by its negative-root equation at a magnitude at least $12a$. Its nonpositive-block arrow with diagonal $-20a$ has increment at most $L_1 = e_-$: a larger positive root would require a sum greater than $20a$, contradicting the test. Adding gives root diagonal $-32a$, with increment at most $4m + e_+ + e_-$. By (8.2) the increase over d is at most $3m$. Larger β only decrease positive trace. This proves every case and the application to the source residues. □

8.1 Parameter carry-through, without changing analytic hypotheses

The following audit concerns actual occurrences of the constants, in addition to reading the full proofs. It is not a replacement for those analytic proofs.

1. In the finite section, stability, quotient representation and residue averaging do not require $\beta = 80$. The support bound becomes $(1 + 32t)H_0$, the second-moment bound becomes $((32t)^2 + K)W_v^2$, the initial root threshold remains $3/\sqrt{r}$, and norm detection becomes $64t + \alpha$. The only lower restriction on the buffer comes from the assignment proof, now supplied by Theorem 8.2.

2. In `invariant.tex`, β enters the bounded support, shifted quotient, and permitted fixed-parameter constants. The analytic off-diagonal perturbation is independent of the size of the diagonal shift. No inequality uses $\beta \geq 80$. Bounded costs, ordered integral changes and rank comparison have the same hypotheses with the fixed buffer 32.

3. In `coloring.tex`, the lift, joint cutoffs, second-moment tightness, argument recovery, height homogeneity and strip locality require a fixed finite buffer, not 80. The bad-ball threshold is $2\beta t + \alpha$. The assignment bound passes to the finite-cut limits with 9 in place of 24. The hazard recurrence and its finite iteration become

$$\mathcal{S}(c^h) \leq (1+9h)\mathcal{S}(c) + hg(h), \qquad S_N \leq \frac{g(h)}{9}((1+9h)^N - 1). \tag{8.5}$$

   For fixed T_0 and $N = \lceil T_0/h \rceil$, the same limit argument works since $(1+9h)^N \leq e^{9(T_0+h)}$. Fresh times, masks, parameterized L^1 limits and random tie symmetry are unchanged; their moduli may change with the fixed parameters.

4. The kernel theorem, Bernoulli free product and fixed-word transfer have no occurrence of the buffer or assignment constant. In completion only the Cartan numerical estimate changes. The multiplier algebra, clipping scale, fixed-moment order and final spectral cut are unchanged. The expectation-free reduction and corner palette are also unchanged.

Theorem 8.3 (Conditional quantitative paving corollary)**.** *Assume the inherited analytic and structural inputs used by the pinned source: the invariant bounded-cost and rank-comparison statements and their joint-cutoff/homogeneous-cost consequences with fixed buffer 32; the atomless-kernel and Bernoulli free-product constructions; fixed scalar moment transfer and multiplier calculus; and the faithful-state and expectation-free reductions described in Section 2. Retain their stated standard-space, bounded-degree, cocycle-ratio, faithful-state and expectation hypotheses at the stages where they are required. With Theorem 8.2 in place of the source assignment lemma, every inclusion and self-adjoint operator in Section 1 then satisfies* (1.2) *with*

$$r'_\varepsilon = \lceil 9000000\varepsilon^{-2} \rceil + \lceil 4/\varepsilon \rceil + 1 \leq 9000007\varepsilon^{-2} < 10000000\varepsilon^{-2}. \tag{8.6}$$

The number still precedes the inclusion and finite-vector tests. This is a conditional improvement in the source chain, not an independent proof of its inherited inputs.

Proof. The first three items of the carry-through audit, the inherited analytic inputs, and Theorem 8.2 give the source's Bernoulli coloring conclusion with $\beta = 32$, still with exact conditional color marginals $1/r$. The all-unassigned cost is zero, the recurrence (8.5) gives arbitrarily small final cost at fixed total assignment time, and the unchanged bad-ball and completion arguments give the local norm bound $64t + \alpha$ for a contraction.

On the faithful central-state corner take $r = \lceil 3000^2\varepsilon^{-2} \rceil$ and $t = 4/\sqrt{r} > 3/\sqrt{r}$. Apply coloring to $y_n/2$ with $\alpha = \varepsilon/32$, radius n, and failure probability $(\varepsilon/16)^{2n}$. Restoring the factor two gives

$$2(64t + \alpha) = \frac{512}{\sqrt{r}} + \frac{\varepsilon}{16} \leq \left(\frac{512}{3000} + \frac{1}{16}\right)\varepsilon = \frac{1399}{6000}\varepsilon < \frac{\varepsilon}{4}. \tag{8.7}$$

The free compression contribution obeys, since $0 < \varepsilon < 1$,

$$\frac{2}{r} + \frac{4}{\sqrt{r}} \leq \left(\frac{2}{9000000} + \frac{4}{3000}\right)\varepsilon = \frac{6001}{4500000}\varepsilon < \frac{\varepsilon}{4}. \tag{8.8}$$

Hence the source's high-moment estimate, clipping at $\varepsilon/2$, and approximation of Z_n by Q_n of norm at most $3\varepsilon/4$ apply without other numerical changes. For each $\rho > 0$, choose k with $(3/4)^{2k} < \rho/4$, then n, then a cylinder partition, then a fine partition and deterministic evaluation. Moment transfer and the spectral cut give $\varphi(1-q) < \rho$ and compressed norm at most ε on this corner. The inherited expectation-free reduction adds $\lceil 4/\varepsilon \rceil + 1$ colors, with diagonal norm at most one and arbitrary strong tests. Scaling restores $\|x\|$.

Finally $\lceil 9000000\varepsilon^{-2} \rceil + \lceil 4/\varepsilon \rceil + 1 \le 9000000\varepsilon^{-2} + 4\varepsilon^{-1} + 3 \le 9000007\varepsilon^{-2}$, proving (8.6). □

The convenient advertised bound 10^7 is fifty times smaller than the source's advertised $5 \cdot 10^8$, with the same exponent two. This is a numerical consequence of a proved finite step together with explicitly inherited analytic inputs. It is not evidence of a new exponent, an unconditional independent proof, or measured computational acceleration. The internal contribution assessment also relies on the different finite-ball certification results, not on this numerical factor alone.

9 Ram Correspondences and Full Certificates

The actual local passages in Appendix B specify finite spaces, compatible measurements, physical metrics, exact annihilators, and residual propagation. Their use here is more precise than renaming a characteristic polynomial a Ram polynomial.

9.1 Cancellation-aware cyclic moment coordinates

For a given v, compute $d = \gcd(P_I, Q_v)$ monic, and write $P_I = dp$, $Q_v = dq$. Set $k = \deg p$. The spectral representation proves

$$\frac{q(z)}{p(z)} = \sum_{j=1}^{k} \frac{w_j}{z - \lambda_j}, \qquad w_j = \frac{q(\lambda_j)}{p'(\lambda_j)} > 0, \quad \sum_j w_j = 1,$$

where the roots of p are distinct real numbers. Only the cyclic spectral data are retained by this quotient. The common factor d may contain positive roots and repeated multiplicities invisible at the root.

Proposition 9.1 (Finite physical moment pencil)**.** *Let $\mu_j = \sum_h w_h \lambda_h^j$, $Jc = \sum_{j=0}^{k-1} c_j H_v^j e_0$, and*

$$\mathsf{H} = (\mu_{i+j})_{i,j=0}^{k-1}, \qquad \mathsf{L} = (\mu_{i+j+1})_{i,j=0}^{k-1}, \qquad C = \mathsf{H}^{-1}\mathsf{L}.$$

Then $\mathsf{H} > 0$, $J^*J = \mathsf{H}$, $H_v J = JC$, $\mathsf{H}C = C^*\mathsf{H}$, and

$$\det(z\mathsf{H} - \mathsf{L}) = \det(\mathsf{H})p(z), \qquad e_0^*\mathsf{H}(zI - C)^{-1}e_0 = G_v(z). \tag{9.1}$$

The moments through degree $2k$ uniquely determine ρ_v among positive real measures, but do not determine d.

Proof. For a coefficient vector c the Gram form is $\sum_h w_h |\sum_j c_j \lambda_h^j|^2$. A polynomial of degree below k vanishing at k distinct points is zero, so the Gram matrix is positive definite. The cyclic space is invariant because $p(H_v)e_0 = 0$. Compression is therefore exact: $H_v J = JC$. This proves the metric symmetry, determinant and resolvent identities. The extended Hankel matrix through degree $2k$ annihilates the coefficient vector of p. Any other positive representing measure must have $\int |p|^2 = 0$, hence be supported on its zeros. The first k moment equations then uniquely determine the weights by the Vandermonde matrix. Multiplying numerator and denominator by a common real-rooted polynomial leaves every rooted moment unchanged, proving the last limitation. □

This realizes volume 21's `v21c02:zero-energy` and `v21c02:atomic-certificate` on the actual color–sign quotient. A candidate certificate must check exact gcd, real root isolation, positive weights, monic normalization, and reproduction of every supplied moment. PSD of a few Hankel/localizer matrices is not sufficient by itself for an arbitrary truncated moment problem. No small singular value is rounded to an exact annihilator. For a local ball in Theorem 6.2, this pencil certifies that ball's rooted measure; it does not reconstruct the entire infinite-graph spectral measure from finitely many moments.

9.2 Transport the observation and metric together

TR250's `eq:moments` prescribes a finite measurement map $m = Ec$ and an inverse on the compatible range. Here the physical space is the finite invariant cyclic subspace $\operatorname{ran} J \subset \mathbb{C}^{|I|}$, with all functionals bounded and no hidden boundary traces. For a square invertible E, define

$$C_E = ECE^{-1}, \qquad \mathsf{H}_E = E^{-*}\mathsf{H}E^{-1}, \qquad e_E = Ee_0.$$

Multiplication gives $\mathsf{H}_E C_E = C_E^* \mathsf{H}_E$ and

$$e_E^* \mathsf{H}_E (zI - C_E)^{-1} e_E = G_v(z).$$

The root vector must be transformed: preserving the matrix spectrum while forgetting its observation changes the rooted cost. A rectangular bank needs full column rank, a specified left inverse and compatible-range/noise treatment. Invertibility alone supplies no conditioning margin. This is compatible coordinate transfer in the private representation sense, not a new spectral theorem.

For a smaller injective synthesis J_0, set $H_0 = J_0^* J_0$, $P_0 = J_0 H_0^{-1} J_0^*$, $C_0 = H_0^{-1} J_0^* H_v J_0$. The physical-metric passage in volume 21 gives, by orthogonal decomposition,

$$\|(H_v - \lambda) J_0 c\|^2 = \|(C_0 - \lambda)c\|_{H_0}^2 + \|(1 - P_0) H_v J_0 c\|^2 . \tag{9.2}$$

The omitted term vanishes for the exact cyclic space, not for an arbitrary moment compression. Relative metric inequalities $aH_0 \preceq \widehat{H}_0 \preceq bH_0$ convert physical norms with factors $\sqrt{a}, \sqrt{b}$. Entrywise closeness alone need not do so.

9.3 Schur residuals, polynomial filters, and local uncertainty

Core report 5's `prop:schurresidual` applies to $zI - H_v$ with $\Im z = \eta > 0$. For an approximate solution $(\widehat{u}, \widehat{w})$ of $(zI - H_v)(u, w) = (1, 0)$ define

$$r_0 = 1 - (z + a_v)\widehat{u} + g_v^* \widehat{w}, \qquad r_I = g_v \widehat{u} - (zI - \Lambda_v)\widehat{w}.$$

Eliminating $\widehat{w}$ gives exactly

$$1 - F_v(z)\widehat{u} = r_0 + g_v^* (zI - \Lambda_v)^{-1} r_I, \qquad |\widehat{u} - G_v(z)| \le \frac{|r_0|}{\eta} + \frac{\|g_v\| \, \|r_I\|}{\eta^2}. \tag{9.3}$$

The bound uses both Hermitian resolvent estimates $|G_v(z)| \le 1/\eta$ and $\|(zI - \Lambda_v)^{-1}\| \le 1/\eta$. The full residual also gives $|\widehat{u} - G_v| \le \|(r_0, r_I)\| / \eta$. A small interface residual without the interior term is not an error certificate.

The same report's `eq:filtererror` requires a certified spectral interval and uniform polynomial error. Lemma 6.1 supplies that error for the particular positive-part filter, and separately the derivative error needed by a deletion increment. If $B_N = \sum_j b_j x^j$ and computed moment intervals

have radii ϵ_j, an additional $\sum_j |b_j|\epsilon_j$ bounds the integral's arithmetic uncertainty. Changing to an orthogonal moment bank requires transforming its uncertainty set as well as its coordinates. The Woodbury identity can reuse same-space arrow solves after a rank-two root change, but requires a certified invertible middle matrix and full residual budget. Neither reuse nor speedup was implemented or measured here.

10 Verification and Source-Specific Examples

The accompanying bounded exact-arithmetic program compares independently constructed outcome averages, determinant differentiation, cycle expansion, deletion identities, and conditional laws. It also tests the local formulas against larger principal sets. Executed results, versions, source hashes and rendering evidence are kept in local audit records. Finite tests supplement the proofs; they do not prove dimension-uniform stability or the global theorem.

10.1 Two sites distinguish all three notions

Take $T = \left(\begin{smallmatrix} 0 & 1 \\ 1 & 0 \end{smallmatrix}\right)$, weights one, both sites unassigned, $r = 2$, and $t = 1/10$. Then

$$P(z) = (z + 1/10)^2 - 1/4, \qquad Q_v = z + 1/10, \qquad \rho_v = \tfrac{1}{2}\delta_{2/5} + \tfrac{1}{2}\delta_{-3/5}.$$

Thus $m_v = 1/5$, $d_v = 2/5$, and $s(P) = 2/5$. A pair collision occurs with probability $1/4$; only then does an outcome have a positive eigenvalue, of size $9/10$. Consequently

$$\mathbb{E}\,\mathrm{Tr}(B_\omega)_+ = 9/40 \neq s(P) = 2/5.$$

Without the independent signs, the color-only polynomial would instead have constant correction $-1/2$, not $-1/4$. This is a direct diagnostic for an accidental use of M594's law. No initial-cost theorem is invoked at this small t, which does not satisfy $t > 3/\sqrt{r}$.

10.2 An invisible repeated root in admissible data

Take three sites with only $T_{12} = T_{21} = 1/2$ nonzero, weights one, all assigned color one, $r = 1$, buffer $\beta = 80$, and $t = 1/80$, so all shifts are one. At the isolated root 3,

$$P(z) = (z + 1)((z + 1)^2 - 1/8), \qquad Q_3(z) = (z + 1)^2 - 1/8, \qquad G_3(z) = 1/(z + 1).$$

The root measure is just δ_{-1}; the other two roots of P are invisible there. At a root in the edge component the isolated factor $z + 1$ cancels instead. Duplicating isolated vertices produces repeated full characteristic factors without increasing the observable rank. This illustrates why the full polynomial pair and its gcd remain part of a certificate.

10.3 An exact local approximation with known error

For the two-site example with buffer $\beta = 80$, use the valid common support bound $L = 9$ from $\beta t = 8$, including the permanent options, even though the parent spectrum is much smaller. At $N = 2$,

$$B_2(x) = \frac{(x + 9)^2}{36}, \qquad M_2 = \frac{\mu_2 + 18\mu_1 + 81}{36}, \qquad e_2 = 9/\sqrt{2}.$$

The certificate is conservative; a large inherited support interval can make its lower bound zero. For a parent-only calculation one may use its tighter certified interval $[-11/10, 11/10]$. Using the tighter parent support for a permanently shifted child would be invalid. The degree and support tradeoff is explicit in (6.4), not hidden in an unreported computation.

10.4 Check contract

Exact tests are finite and bounded: rational and complex-rational Hermitian matrices of size at most five, color counts at most three, and path-locality tests with specified maximum size and degree. The required invariants are the normalized differential identity, all site deletions, the even-cycle factors, root-option averaging and shift identities, the sign-aware second moment, forest realization, and moment/deletion-increment locality. The sharpness path is tested inside the source's contraction and positive-shift hypotheses. Arithmetic comparisons of a few roots are not used to infer a general real-rootedness theorem.

The executed suite passed all 16 recorded groups under Python 3.10.9, SymPy 1.11.1 and NumPy 1.23.5. Seven base color–sign fixtures gave 19 deletion checks and 26 root-option checks; four additional buffer-32 fixtures brought the direct outcome enumeration to 454 outcomes. Four forest fixtures passed. Larger-set comparisons gave 405 exact rooted-moment equalities and 405 exact deletion-increment equalities through degree eight. Six sharpness paths used at most seven vertices and degree twelve. Exact Hankel and coordinate-transfer checks, Bernstein bounds at degrees $2, 4, 8, 16$, and rational arithmetic for both completion bounds also passed. The factor-two selection test passed 7,380 exhaustive pairs of nonnegative arrays with entries in $\{0, 1, 2\}$ and lengths at most four.

Separately, 96 seeded floating-point arrow diagnostics tested the buffer-32 inequality, including 52 cases with $m \geq a$ and 44 with $m < a$, with tolerance 10^{-10}. They passed but are not interval certificates or a proof of a universal inequality; Theorem 8.2 supplies that proof. The report makes no empirical runtime or speedup claim. No code evaluates an unbounded family of graphs, performs an unbounded symbolic search, or substitutes a small test for a measured convergence theorem.

11 Comparison, Limits, and Research

Realized local gains. The sign-sensitive cycle formula gives a checkable finite model different from color-only pinching. On forests it yields a single sparse Hermitian realization valid for every principal restriction. The mixed-polynomial half-radius theorem reduces the exact information needed for rooted moments and polynomial deletion increments. The Bernstein derivative calculation supplies an explicit, dimension-independent cost tolerance on that smaller ball, and the selection corollary propagates it to a rigorous one-step assignment budget. The buffer-32, cost-nine theorem is fully proved locally, and Theorem 8.3 traces its improved global constant with all inherited analytic dependencies explicit. These are gains beyond generic coordinate relabeling; they do not independently establish the existence or efficiency of a globally coherent coloring.

What is classical and what is represented. Permutation expansions, formal logarithms, Euler tours, matching determinants, Schur complements, spectral measures, Hankel/Vandermonde factorizations, and Bernstein approximation are classical techniques. The private Ram texts prescribe compatible finite measurements, full physical metrics, exact atomic verification and residual accounting. They are used here to preserve the operator, observation, positivity and uncertainty under representation. None of these uses establishes a historically new discovery. The source already contains real stability, arrow realization, rooted mass allocation and the assignment inequality; they are not credited as this report's improvements.

The public literature separates finite multi-paving from arbitrary-MASA strong paving. Ravichandran–Srivastava's mixed-determinantal theorem supplies the finite upper bound [2]. Popa–Vaes formulates the quadratic arbitrary-MASA conjecture and the expected-ultrapower transfer [3]; the optimal singular-MASA estimate is a distinct theorem with distinct hypotheses [4].

Leake–Ravichandran develops related mixed-determinant methods [5]. These public inputs were checked by primary title/topic searches on the report date; the search is not evidence that no newer relevant result exists.

Advantages and costs. The finite certificate has an explicit radius and error, preserves both signs and all deletion conventions, and remains meaningful at multiple roots after exact cancellation. On forests it preserves graph sparsity. However, cycles prevent a general sparse matrix replacement of the same kind; determinant coefficients can grow; monomial moments and exact root isolation may be ill-conditioned or expensive; and polynomial enclosures can be conservative. Exact rank or residue claims require exact arithmetic or certified enclosures. A list of positive moments does not identify a full characteristic factor invisible to the root.

Unmet obligations. The radius depends on the largest weight and therefore does not replace the source's high-deletion lemma, joint-cutoff uniformity, homogeneous cost construction, or hazard conditioning. One finite approximate color choice does not preserve an entire measured process without a summable error analysis and equivariance. The type I normalizer argument, atomless-kernel theorem, fixed-word nontracial transfer and multiplier clipping remain separate structural work. The report improves the universal color constant only conditional on those inherited inputs, does not improve the exponent two, and does not independently validate the global source theorem.

Focused research targets. First, combine the explicit bounded-ball certificate with a quantitative upper/lower cutoff modulus; at present the source's analytic compactness step is still non-effective here. Second, exploit cycle structure beyond forests, keeping the full even-cycle coefficients rather than a pairwise collision surrogate. Third, test physically scaled orthogonal moment banks with interval arithmetic and record total arithmetic cost before asserting speed. Fourth, propagate certified local assignment errors through the hazard recurrence while preserving random tie symmetry. A successful treatment of any of these must retain the order of quantifiers in (1.1)–(1.2).

A Source Scope and Input Provenance

The local source root is

> `C:/Users/mural/OneDrive/Documents/OpenAI-722-math/_sources/openai-math/preprints/Quadratic-Strong-Operator-Paving-over-Arbitrary-Maximal-Abelian-Subalgebras-September-25-2026/build/source/`

Read in full: `main.tex`; `sections/introduction.tex`; `sections/reductions.tex`; `sections/kernels.tex`; `sections/polynomials.tex`; `sections/invariant.tex`; `sections/coloring.tex`; `sections/transfer.tex`; `sections/completion.tex`; `references.tex`; and `figures/shortest-pair.tex`. All mathematical proofs, conditional statements, figure interpretation and references are included in this scope; no additional mathematical appendix is present. The catalogue row was read at the stated commit pin. SHA-256 hashes, line counts, command results and the actual source-read scope are recorded locally, separately from the mathematical publication.

For context, the completed report `M594-MASA-Paving/TR-M594-Ram-MASA-Paving.tex` was read with its verification record. Its deterministic half-radius result motivated a question, not an assumed formula for the present color–sign model. Every M595 source file listed above was read independently.

B Exact Private Corpus Passages

The following local paths are confidential. Their mathematical roles are stated in Section 9; none is treated as a proof of arbitrary-MASA paving.

1. `C:/Users/mural/OneDrive/Documents/RamFoundationTRs/TR-rev-1-old-tr250/TR-rev-1-old-tr250.tex`. Read lines 121–274: *Spaces, operator domains, and Ram coordinates*, *What a Ram representation means in this report* with `eq:moments`, and *Ram moment coordinates that retain the complex* with `thm:transfer`; also lines 455–477, *A concrete polynomial Ram construction*. These distinguish finite compatible coordinate transfer from continuum approximation and require metric and domain preservation.

2. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch02/finite.tex`. Complete file: *What a Moment Record Specifies*, *The polynomial functional*, `v21c02:hankel`, `v21c02:localizer`, `v21c02:zero-energy`; *Finite Atomic Recovery With a Complete Check*, `v21c02:atomic-certificate`; its three-point example and *Finite Nullspaces and Identified Observables*. Positivity, support, identification and stability are different claims.

3. `C:/Users/mural/OneDrive/Documents/RamCalBooks/volumes/volume21/chapters/ch01d-physical-metrics.tex`. Complete file, especially *A Nonorthogonal Representation Carries a Metric*, `v21c01:physical-gram-definition`, `v21c01:metric-symmetry`; *The Full Residual Has an Omitted Component*, `v21c01:residual-decomposition`, `v21c01:residual-identity`; and *Approximating the Metric*, `v21c01:gram-equivalence`. Its final finite-measurement warning is retained.

4. `C:/Users/mural/OneDrive/Documents/RamPatsCore50-TRs/TR-core-pat-5/TR-core-pat-5.tex`. Read lines 281–438, including *Sampling and discarded subspaces*; *Polynomial evaluation without eigendecomposition*, `eq:filtererror`; *Coarsening, Lifts, and Schur Sources*, `prop:schurresidual`, `eq:schurresidual`; and *Topology Edits and Checked Low-Rank Reuse*, `thm:woodbury`, through *Middle conditioning is necessary but not enough*. The full-space and residual requirements, not generic graph terminology, are used here.

References

[1] OpenAI, *Quadratic Strong-Operator Paving over Arbitrary Maximal Abelian Subalgebras*, September 25, 2026, catalogue M595, commit `adc7f1241b42e322a6451854ab7e4b4c146bf78a`. Pinned public manuscript; local complete source in Appendix A.

[2] M. Ravichandran and N. Srivastava, *Asymptotically optimal multi-paving*, International Mathematics Research Notices **2021** (14), 10908–10940. Definition 5, Theorem 15, Proposition 16 and Theorem 25 in arXiv:1706.03737v2. https://doi.org/10.1093/imrn/rnz111.

[3] S. Popa and S. Vaes, *Paving over arbitrary MASAs in von Neumann algebras*, Analysis & PDE **8** (2015), 1001–1023. Definition 2.2, Theorem 2.7 and Conjecture 2.8. https://msp.org/apde/2015/8-4/apde-v8-n4-p06-s.pdf.

[4] S. Popa and S. Vaes, *On the optimal paving over MASAs in von Neumann algebras*, Contemporary Mathematics **671** (2016), 199–208. https://arxiv.org/abs/1507.01072.

[5] J. Leake and M. Ravichandran, *Mixed determinants and the Kadison–Singer problem*, Mathematische Annalen **377** (2020), 511–541. https://arxiv.org/html/1609.04195v2.

[6] M. Subbarao, confidential Ram Calculus corpus, exact local paths, headings and labels in Appendix B. Used as representation and certificate contracts, not public evidence of historical novelty.